Array

Presented By : Mazen Othman

- Problem Solving Instructor

Agenda

- Introduction to Array
- One Dimensional Array
  - Dynamic Array
    - Insert Element To Array
    - Display Element From Array
  - Access Elements & Memory Location >
  - Add & Update Elements
- Two Dimensional Array
- Class Array
  - Difference Between C-Style Array & Class Array
- Bonus Question

Array

Given 3 numbers, print their sum, then print each element.

```
int a1 ,  a2 , a3 ;
cin >> a1 >> a2 >> a3;
cout << a1 + a2 + a3 << endl ;
 cout << a1 << ‘ ’ << a2 << ‘ ’ << a3;
```

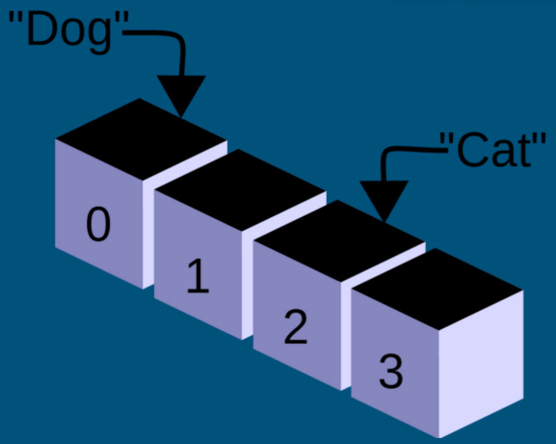
What about 10 numbers?

```
int a1, a2, a3, a4, a5, a6, a7, a8, a9, a10;
```

What about 1000 numbers ?

Array

Arrays are used to store multiple variables in same data type in a single container, instead of declaring separate variables for each value.



Array

To declare an array, define the variable type, specify the name of the array followed by square brackets and specify the size of array it should store

```
int nums [2] = {1, 3} ;
```

Variable type

Name of array

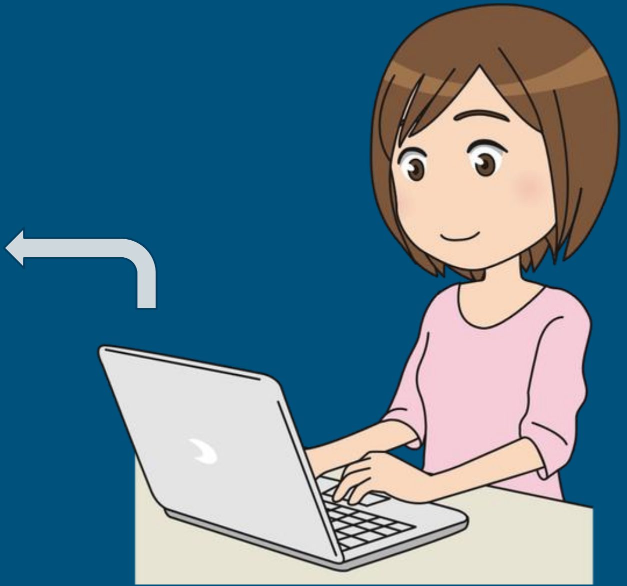
Size of array
(in square brackets)

elements

Array

- **Array Declare Methods :**

```
//First method :
int arr[5] = {1,2,3,4,5};
int arr[5] = {1,2};
int arr[] = {1,2,5,2};
//Second method :
int arr[3];
arr[0] = 1;
arr[1] = 2;
arr[2] = 3;
/*That will make us an array with size the that
contains the following elements [1,2,3].
*/
```



Access Element

Example :

```
int nums [5] = {1, 3, 6, 2, 9};
```

	1	3	6	2	9
Indexes →	0	1	2	3	4

Add & Update Elements

Example :

```cpp
#include <iostream>
using namespace std;

int main() {
 // creating an array
  string names[3] = {"Theo", "Ben", "Dalu"};

  // changing the first element of the array
  names[0] = "David";

  // printing the new elwment
  cout << names[0];
}
```

Array

- **So We Can Say That :**

Any operation that we can do on a normal variable we can do on an array.

For example if we have int arr[3] = {1,2,3};

We can : arr[0] = arr[1] + arr[2] ;

That will make the first element = 2 + 3 = 5, the array will be {5,1,3}

We can : cin >> arr[2] ;

that will replace the last element with the user input

We can : cout << arr[1];

that will output the second element of the array

Class Array VS C-Style Array

```cpp
#include <bits/stdc++.h>

using namespace std;

int main() {

    array <int,5> Points ={1,3,6,2,9};
}
```

```cpp
#include <bits/stdc++.h>

using namespace std;

int main() {

    int Points[5] ={1,3,6,2,9};
}
```

Note

" #include <bits/stdc++.h "

it is include all the

standard libraries.

Array

- ## Cin & Cout In Array

We can use loops with arrays so for example if we want to take 100 numbers from the user and store them we will use a for loop.

int numbers[100];

```
for( int i = 0 ; i < 100 ; i++ ){
cin >> arr[i];
}
```

Array

- **Cout In Array**

```
for( int i = 0 ; i < 100 ; i++ ) {
cout << arr[i] ;
}
```

Displaying the array

Two Dimensional

Graphic Representation for Two Dimensional Array 3*3

	Column 0	Column 1	Column 2
Row 0	x[0][0]	x[0][1]	x[0][2]
Row 1	x[1][0]	x[1][1]	x[1][2]
Row 2	x[2][0]	x[2][1]	x[2][2]

Graphical Representation of Two-Dimensional Array of Size 3 x 3

Two Dimensional

Syntax for Two Dimensional Array

❖ Initialization of Two-Dimensional Arrays in C++

The various ways in which a 2D array can be initialized are as follows :

1. Using Initializer List
2. Using Loops

❖ Initialization of 2D array using Initializer List :

We can initialize a 2D array in C by using an initializer list as shown in the example

.

Two Dimensional

Syntax for Two Dimensional Array

This array has 3 rows and 4 columns. The elements in the braces from left to right are stored in the table also from left to right. The elements will be filled in the array in order: the first 4 elements from the left will be filled in the first row, the next 4 elements in the second row, and so on.

```
//First Method:

int x[3][4] = {0, 1 ,2 ,3 ,4 , 5 , 6 , 7 , 8 , 9 , 10 , 11}
```

Two Dimensional

Syntax for Two Dimensional Array

This type of initialization makes use of nested braces. Each set of inner braces represents one row. In the above example, there is a total of three rows so there are three sets of inner braces. The advantage of this method is that it is easier to understand.

```
//Second Method ( better ):

int x[3][4] = {{0,1,2,3}, {4,5,6,7}, {8,9,10,11}};
```

Two Dimensional

Syntax for Two Dimensional Array

❖ Initialization of 2D array using Loops

We can use any C++ loop to initialize each member of a 2D array one by one as shown in the example.

```
int arr[3][4];

for(int i = 0; i < 3; i++){
    for(int j = 0; j < 4; j++){
        cin >> arr[i][j];
    }
}
```

Two Dimensional

Syntax for Two Dimensional Array

❖ Display Elements of 2D array using Loops

We can use any C++ loop to Display each member of a 2D array one by one as shown in the example.

```
int arr[3][4];

for(int i = 0; i < 3; i++){
    for(int j = 0; j < 4; j++){
        cout << arr[i][j];
    }
}
```

Task Question

- If We Give An Size To Array And Don't Use What Will Happen And Why ?

Bonus Question

- **Problem Statement**

- Masterpiece Problem
- 2D Problem
- Masterpiece Problem 2

Any Questions

Thank You